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FEATURE ARTICLE

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Software as a service: Lessons from the video game industry

Fanny Vaudour | **Aleksej Heinze**

Digital transformation has prompted some organizations to use a service business model that does not require the purchase of a product but, rather, use of their services. Software-as-a-service (SaaS) is a method by which service providers meet the needs of their clients via an internet server. A look at consumers' reactions to the games-as-a-service (GaaS) business model used in the computer games industry illustrates how the model has transformed the industry. It also highlights the importance of timeliness, transparency, and creativity in any SaaS business's efforts to attract and retain customers and optimize profitability.

1 | INTRODUCTION

A business model describes how an organization creates, delivers, and captures value (Osterwalder & Pigneur, 2010). Software-as-a-service (SaaS) is a method used by service providers to distribute software-based services to clients via a single server, which may or may not be owned by a third party. Some prominent examples of SaaS are Microsoft Office 365, Google Apps, Dropbox, and Amazon Web Services. Concentrating the provision of software on a centralized server guarantees synchronous distribution, saves time and money, and eases the problems of scalability and compatibility that occur when software is distributed via physical media, such as CDs. SaaS is one of the main applications of cloud computing, and business models based on SaaS have been adopted by several industries—most notably, companies that develop and publish computer games.

The provision of multiplayer and standalone games via the internet—essentially a variant of SaaS—has led to the growth of a distinct games-as-a-service (GaaS) business model, where all the software needed to play the game is located on an external server (Ali, Sultan, Ghani, & Zulzalil, 2019). Although GaaS business models have become increasingly popular in the games industry, the topic is relatively new and deserves further study (Choi & Chen, 2019).

One of the first GaaS video games, World of Warcraft, was released in 2004. GaaS-based video games became very popular in the following years, and sales have continued to grow. Video game revenues in the United States have nearly doubled, from USD 17.5 billion in 2010 to USD 29.1 billion in 2017. Compared to the sales of physical media, digital sales—including subscriptions, full digital games, downloadable content, mobile apps, and social network games—have more than doubled. Gaming software delivered via physical media, which accounted for two-thirds of sales in 2010, has now largely been replaced by digital delivery (Entertainment Software Association, 2018).

> *"The provision of multiplayer and standalone games via the internet –essentially a variant of SaaS –has led to the growth of a distinct GaaS business model."*

Gaming is a fast-moving and highly competitive industry. Games developers need to keep track of the rapidly changing trends in the market if they are to keep their audience engaged, and produce games that will keep them coming back for more. The SaaS approach

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means that customers can try out a new game quickly, often at little or no cost. It also means, however, that those same customers can easily move elsewhere, and will pay for a service only if it provides some form of added value for them. In GaaS gaming, only engaged customers will carry on playing the game, become part of the game's community, and ultimately generate revenue.

An exploration of the GaaS business model identifies some of the factors that make some games work while others fail, how the GaaS business model is monetized, and the ways it delivers value to customers. Besides contributing to the development of a more generalized and comprehensive GaaS business model, the lessons this study reveals can be applied to other industries that adopt the SaaS approach and extrapolated for application to other SaaS business models.

2 | THE GAMES-AS-A-SERVICE BUSINESS MODEL

The main business model initially used in the video games industry initially was "one price, one service." Essentially, players bought a game that could be completed from beginning to end in a matter of days, weeks, or months. When the game was over, there was nothing left to do but replay the same game. The adoption of the SaaS approach to software distribution led to a significant change in players' habits. The instant availability of digitally delivered games meant that players began to move away from traditional boxed products toward online versions. The cost was the same, and they were available for play immediately after being downloaded. Consumers did not have to bother with finding a retailer, or suffer the disappointment of being told that the game they were hoping to purchase had sold out. If they wanted to buy a game at 3:00 a.m. on a Sunday, they could.

Digital delivery also changed the dynamics of the games market in other ways. Previously, developers would create a sequel to a popular game to generate additional revenue. With the GaaS model, they can update a game on the fly—that is, release as many updates as they want, whenever they want—allowing all players to receive them as soon as they connect to the server. With GaaS-based games, players never really finish a game because it keeps evolving.

Developers and publishers can extend the life of a game by continually adding new content—commonly referred to as downloadable content, or DLC—as required. DLCs can take many different forms, and may be free or need to be purchased. Some examples include new skins (changing the appearance of a character), levels and quests (to improve the game's storyline), maps and characters, and gaming modes.

Now conducted year-round, game development has become an ongoing rather than a discrete activity. Game developers and publishers realize, however, that gamers will continue to play the game and remain loyal only for as long as their expectations are met (Sinclair, 2017). Consequently, if a game is to be successful, developers need to keep players engaged and continue to satisfy their demands over periods of months or possibly even years.

3 | HOW IS GAAS MONETIZED?

To be effective as a business model, GaaS must generate revenue. Games developers have implemented several different monetization methods and, as their colleagues in other industries have done, often incorporate more than one technique into a single approach.

Most GaaS business models rely on powerful customer relationship management (CRM) systems to segment users and to follow up on gamers' behavior (NoBlue, 2018). In addition to monitoring gamers' behavior, these systems are also used to manage and approve revenue-generating transactions, such as those described below.

Most GaaS business models rely on powerful customer relationship management.

Microtransactions—These are micropayments made online to purchase virtual goods in a game. Apart from being a source of revenue, microtransactions increase players' engagement by offering them attractive items or in-game features.

For example, one popular form of microtransaction offers users a way to skip parts of the game. Gamers can buy access to new content without having to invest time normally spent in building their game characters or working their way through the lower levels of the game, colloquially referred to as "the grind." Instead of grinding, gamers can reach a certain level directly by purchasing it through a microtransaction. The timing of such microtransactions in the game is critical, however: If they interrupt the gaming process, they can lead to a loss of earnings for the game developer and publisher (Tomić, 2019).

Freemium and advergaming—Microtransactions are widely used in free-to-play games because they provide

a source of revenue for developers. Therefore, free-to-play games are rarely entirely free and might more accurately be called freemium or free-to-start games. Although it is possible for players to grind their way through a game, many will prefer to pay to reach higher levels.

Freemium games may also generate income through advertising. For example, after reaching a certain level, players will be shown an advertisement. Often these advertisements will be used to promote a microtransaction, such as blocking further advertisements or gaining advantages in the game. This practice is particularly popular in mobile games.

Freemium games may also generate income through advertising.

Subscription services—A subscription to a game platform, such as Origin by Electronic Arts (EA), or to a multiplayer online role-playing game, such as World of Warcraft by Activision Blizzard, gives gamers full access to a game for as long as the subscription is active. By paying a monthly fee, gamers will be given access to all the games the platform has to offer. For example, Origin has Origin Access, a subscription program that allows gamers to play any EA game, to gain access to new games before nonsubscription gamers, and to get a discount on microtransactions.

One of the goals of subscription services is to increase the loyalty and engagement of gamers. A purely subscription-based game does not require the user to make any other purchase to download or play it; nevertheless, gamers can use microtransactions to purchase items not included in the subscription (NoBlue, 2018).

Season pass—This gives the purchaser access to all new DLCs for a predetermined period, usually a year. Instead of purchasing each DLC separately, gamers pay upfront for a year's worth of DLCs at a discounted price. In addition to guaranteeing access to new DLCs, publishers usually also offer exclusive in-game items that are available only to gamers who have purchased a season pass. This is another way to retain gamers' engagement and reward their loyalty.

Loot boxes—These contain items that players might use to customize their game, such as changing the appearance of their character or gaining access to certain weapons or accessories. The content of each loot box is based on a random number, however, and buyers have no idea what they will get until after they have paid for it.

One of the goals of subscription services is to increase the loyalty and engagement of gamers.

Gamers sometimes are given a limited time in which to purchase loot boxes. This increases the pressure on players to buy them, although mostly they are simply offered one of the items available from the in-game store. Loot boxes are also offered as a reward for gamers who reach a certain level in the game, or as part of a promotion organized by the publisher.

4 | A BUSINESS MODEL FOR GAAS

As with other business models, the business logic of GaaS can be represented using the business model canvas (Osterwalder & Pigneur, 2010). **Exhibit 1** presents a generic business model for GaaS using the nine building blocks of the business model canvas. In addition to listing the essential elements of the GaaS business model, it highlights the relationships between them.

The *customer segments* represent all the customers for whom the GaaS creates value. In this case, the customers are gamers who play GaaS-based games. For each customer segment, there is a specific *value proposition*. Of course, the main value proposition is the game itself, but in addition, GaaS offers new content that comes with updates and DLCs. GaaS also allows bugs to be fixed and provides a mechanism for feedback and communication between gamers and the development teams.

Distribution channels show the GaaS channels that deliver value to gamers. Digital stores, such as Origin, represent the main distribution channels, but physical stores still play a part as some gamers find them valuable.

The *customer relationships* section shows what kind of relationship GaaS brings and how. Developing the players' engagement is part of this business model, as updates usually take gamers' opinions into account. Gamers can find answers to their questions through a digital personal assistant, or they can search for answers in a frequently asked questions (FAQ) database. Developing communities is extremely important for GaaS business models, as they act as a conduit for feedback to the

EXHIBIT 1 Gaming as a service: Generic business model canvas

Key Partners: • Publishers and editors • External development teams • Video game distributors (physical and/or digital) • External software producers	**Key Activities:** • Efficient software development • Production: marketing, designing, and producing • Platforms: updates, improvements, IT, promotions **Key Resources:** • Physical: production studio, video games stores • Intellectual: CRM, production software • Human: skilled development team • Financial: editor's capital	**Value Proposition:** • Constantly changing gaming experience • New content: new quests, skin, map, characters, skills, weapons, etc. • Fixed bugs • Feedback and communication from editors/teams	**Customer Relationship:** • Digital personal assistance • Self service/FAQ • Communities: forums and feedback • Development of the players' engagement **Distribution Channels:** • Digital stores mainly • Physical stores	**Customer Segments:** • Gamers
Cost Structure: • Production • Customer support • Technical infrastructure			**Revenue Streams:** • Subscription • Season pass • Microtransaction • Advergaming	

game producers and play a vital part in building the players' loyalty.

Revenue stream refers to the different models and mechanisms used for bringing revenue to developers and capturing value. As previously noted, these might include microtransactions, advergaming, subscriptions, season passes, and the use of loot boxes.

The *key resources* section shows which assets are vital for GaaS games. They can be divided into four parts: physical, intellectual, human, and financial. Production studios and computer game stores are key resources: The former is where the game is produced, and the latter distributes the game. To be able to handle the players' data and activity, a powerful CRM is needed; to produce video games, sophisticated production software is needed. None of the GaaS video games would be possible without a skilled development team. Finally, financial resources are needed to create, design, and distribute GaaS games.

Key activities are essential to a high level performance. In the case of GaaS, these are efficient software development, effective production (including design and marketing), and suitable IT infrastructure in the form of the platforms that supply updates as required.

The *key partnerships* section includes all those who participate in the business model. GaaS partnerships can be established between publishers, development teams, game distributors, and software producers. Strong partnerships—sometimes exclusive partnerships—can make an important difference in terms of GaaS games. Those partnerships can have an impact on *cost structure*. New entrants, existing competitors, higher expectations by gamers, substitute business models or activities, and the suppliers can all lead to additional and sometimes unexpected costs.

5 | SURVEY DATA AND ANALYSIS

A survey designed to get deeper insights from gamers about the GaaS business model was administered online in June 2019. The invitation to participate was posted on gamer community pages. Some were specifically related to GaaS communities, while others were social networks concerned with games and gaming. A total of 217 responses were received.

Divided into three sections, the questionnaire was inspired by findings from previous studies. Designed to obtain generic background, the first section asked about the awareness of GaaS and experience of playing GaaS games. If respondents had experience with GaaS games, they were asked to complete section 2, which explored their experience in more detail. Of the respondents, 195 fell into this category. Gamers who did not have GaaS experience were directed to section 3, which contained some general questions related to GaaS and the games industry. The remaining 22 respondents fell into this category.

Responses to the first question in the generic background section, *What kind of GaaS models do you know?* showed that most gamers were familiar with microtransactions

(85%), season passes (80%), and subscriptions (79%), while 7% did not know any of those revenue-generating models, and 5% mentioned other models. The other answers included free-to-play and free trials. Several gamers mentioned advergaming-based games, DLCs, and character lifespan payments, which are actually a form of microtransaction. Similarly, two gamers mentioned monthly subscription and streaming services, both of which fit in the subscription category (**Exhibit 2**).

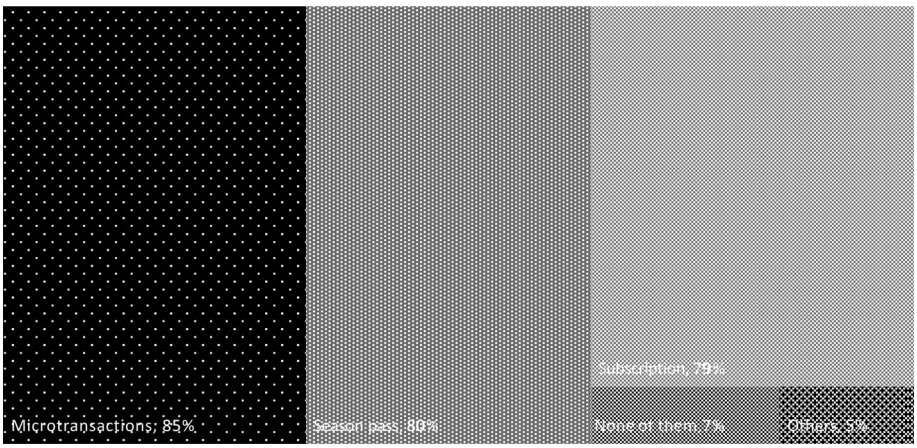
When asked the second general question, *Do you play or have you ever played GaaS video games?* 68% of respondents noted that they were currently playing GaaS games, while 22% had played them in the past but did not play them anymore. Only 10% had never played a GaaS game (**Exhibit 3**).

Those who answered that they play GaaS games or have played them and no longer do so were directed to the second section, where they were asked, *How many GaaS video games have you played in the last 6 months?* More than half (53%) of the respondents indicated that they had played one or two GaaS games in the last 6 months, 22% had played three or four games, and 17% had played more than five games. The remaining 8% had stopped playing GaaS games more than 6 months ago (**Exhibit 4**). These results broadly confirm previous findings on how many GaaS games people can play at the same time without dropping out (Sinclair, 2017).

The next question in this section dealt with the amount of money that gamers are willing to spend on a single game. Of the 195 responses in this section, 105 (54%) were willing to pay up to USD 50, 24% were willing to pay up to USD 100, and 1% were willing to pay up to USD 200. Eleven percent of the gamers would only play free games and 10% did not have a limit on what they would spend (**Exhibit 5**).

These findings show that more than half the respondents want to have the whole gaming experience for USD 50 or less; therefore, the games industry may have to change its business model at some point. The average price for a video game is currently about USD 50 (Loftus, 2019), but adding DLCs, microtransactions, and/or a subscription can add substantially to that sum—which means that players may become ever less inclined to spend additional money on their games.

EXHIBIT 2 What kind of GaaS models do you know?



Note: More than one response was permitted.

EXHIBIT 3 Do you play or have you ever played GaaS video games?

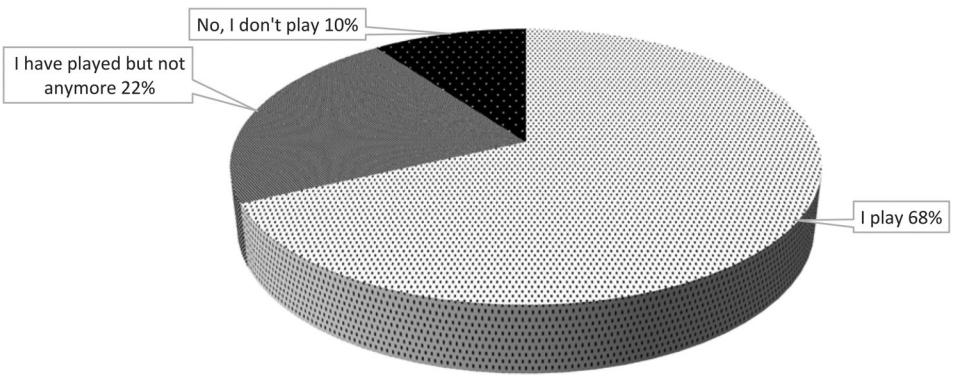


EXHIBIT 4 How many GaaS video games have you played in the last 6 months?

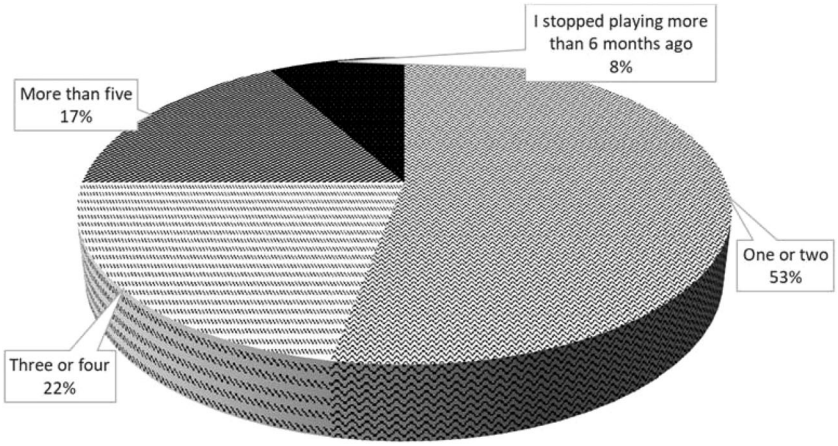
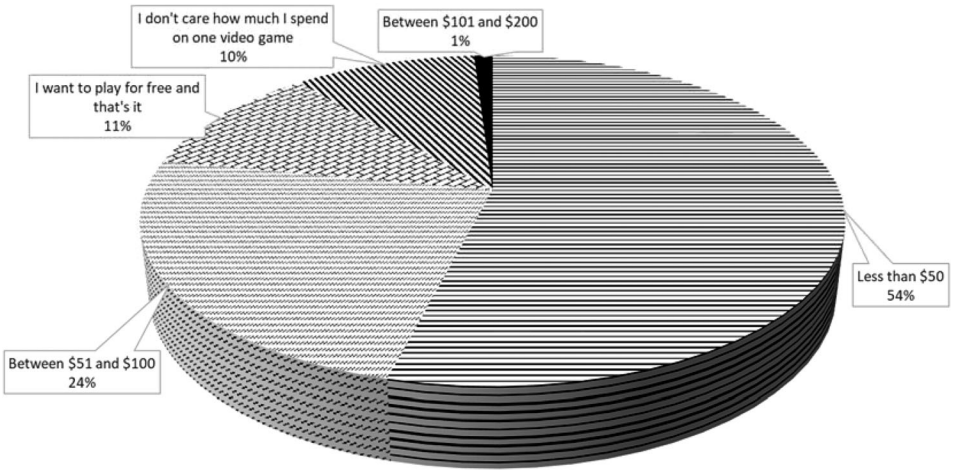


EXHIBIT 5 On average, how much are you willing to spend on one video game?

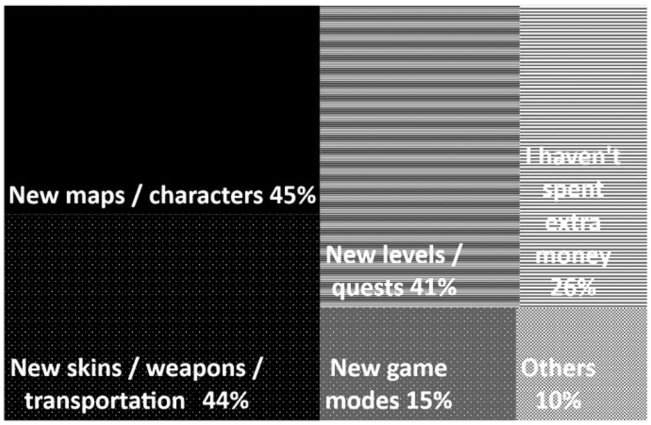


When asked *Have you ever spent extra money on a game once you have bought it? If so, what for?* 50 interviewees (26%) said they had never spent any extra money; 44% had bought new skins, weapons, or transportation for their characters; 41% had bought new levels and/or quests; 45% had bought new maps and/or characters; 15% had spent extra money to have new game modes; and 10% gave another answer (**Exhibit 6**). This question allowed multiple answers to capture the range of purchases that were made.

Of those who answered "Other," several gamers said they were adding a subscription to their game. Others said they were buying DLCs—that is, a package of new game contents, including maps, characters, skins, or weapons. These answers appear to indicate that people have difficulty differentiating between different DLCs and may not necessarily know what they are getting for their money. The situation is further complicated by the fact that in-game currencies can be used to buy loot boxes and other in-game features.

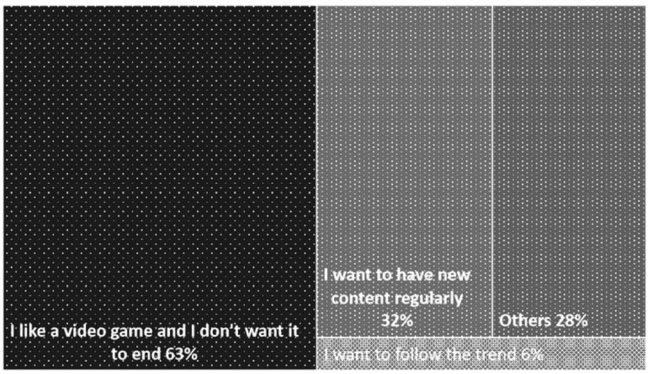
When asked *Why do you play GaaS video games?* more than 63% of the respondents noted that they play because they like a particular game and do not want it to

EXHIBIT 6 Have you ever spent extra money on a game once you have bought it? If so, what for?



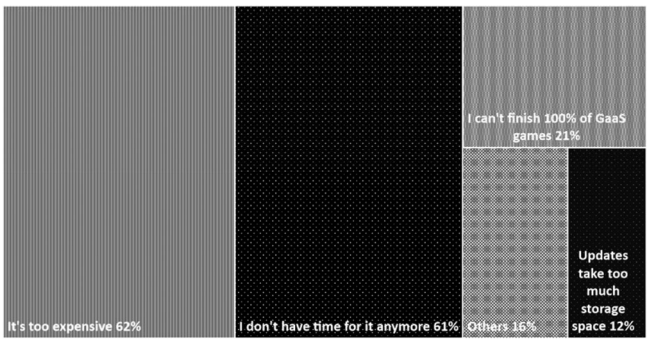
Note: More than one response was permitted.

EXHIBIT 7 Why do you play GaaS video games?



Note: More than one response was permitted.

EXHIBIT 8 What could make you (or what made you) stop playing GaaS video games?



Note: More than one response was permitted.

end, 32% play because their games regularly add new content, and 6% play because they want to follow the latest trends. Quite a high proportion of respondents (28%) supplied their own answer to this question (**Exhibit 7**). Many of the respondents indicated that they were playing GaaS games because almost all the new games had this kind of business model, and they provided a feature that the gamers particularly enjoyed. Several respondents, however, do not find GaaS games appealing and had stopped playing them.

In response to *What could make you (or what made you) stop playing GaaS video games?* cost was the most frequently cited reason (62%), followed by lack of time for it (61%), inability to finish the game (21%), and updates taking too much space on their machines (12%). This question also allowed multiple answers to capture the range of reasons. Previous research has found that gamers with low-speed connections and not enough storage space were not able to enjoy the full gaming experience (Henry, 2017). Sometimes players feel forced to deal with an ever-expanding array of DLCs that they may not want.

Of the 12% of respondents who gave their own answers to this question, several said they had stopped because playing was becoming a question of "grind or pay," and they felt that being required to buy something for a supposedly free game was unethical, a point that echoes previous research findings (Klubnikin, 2018). Similarly, a number of players noted a trend toward "pay-to-win" behavior, which made them feel as though they had been cheated (**Exhibit 8**).

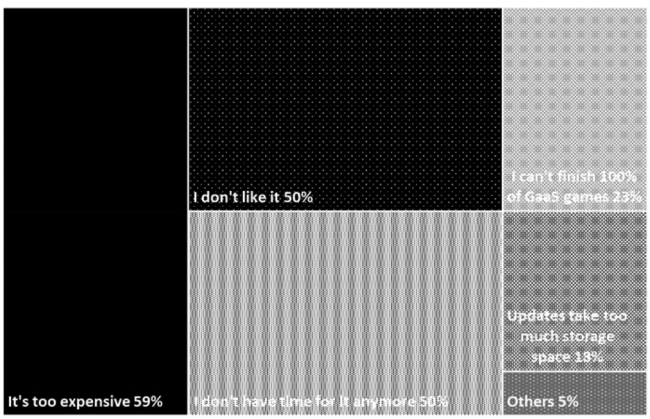
The third section of the survey was for the 22 respondents who indicated that they do not play GaaS video games. Allowing multiple answers, this question attracted two prominent responses related to cost and time. The high cost of the games was most often cited (by 59%) as a reason for not playing those games, followed by not having enough time (50%), inability to complete them (23%), dislike of them (23%), use of too much space on their computer (18%), and other reasons (5%). Only one gamer said that the video games he played did not have a GaaS business model (**Exhibit 9**).

An optional question addressed new trends in the video game industry, and only eight gamers responded. Several noted that the video games tended to be incomplete at launch and were fixed later with DLCs or updates, only some of which were free. The general feeling was that this tarnished the gaming experience.

The last question of this section, *What kind of gamer are you?* used classifications from previous studies (**Exhibit 10**). The results were super gamer (54%), console warrior (11%), transitional gamer (14%), daily dabbler (19%), and incidental player (2%).

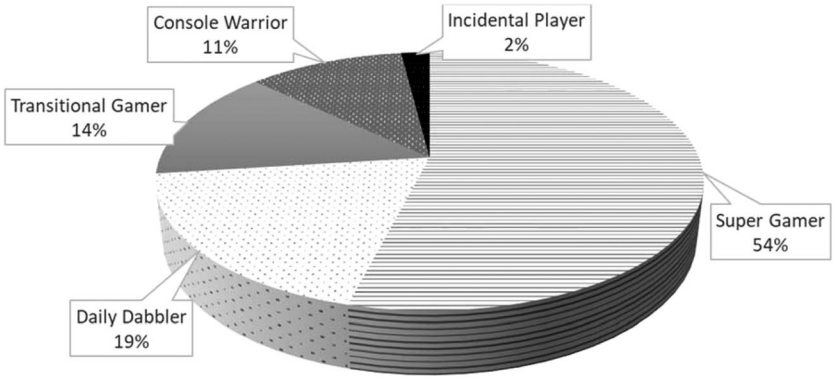
The final survey question, *What do you think about GaaS video games?* asked respondents to provide a rating of 0 (poor) to 10 (great). The two most frequently given ratings were 5 and 7, for an average score of 5.32 (**Exhibit 11**).

EXHIBIT 9 Why don't you play GaaS video games?



Note: More than one response was permitted.

EXHIBIT 10 What kind of gamer are you?



6 | THE IMPACT OF GAAS ON THE GAMES INDUSTRY

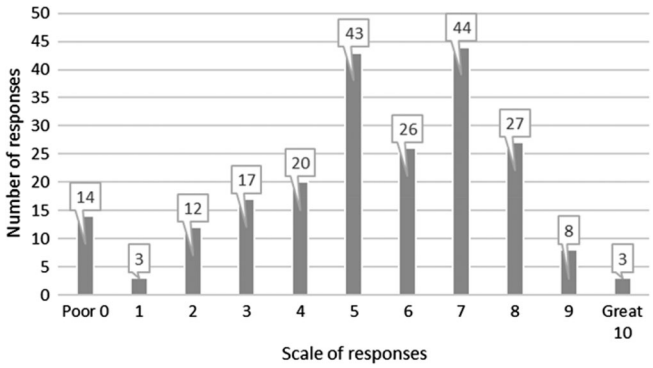
GaaS has revolutionized the games industry for developers, publishers, and players alike. It is clearly a useful model for developers, as they can launch an unfinished game and, if it is well received, improve on it later using updates and DLCs. They do not need to make a large upfront investment in communication and marketing to make a game a success. Once the game is launched, the CRM systems that form part of the GaaS approach make it easier to listen to the community for feedback, and adapt updates accordingly (Sinclair, 2017).

The business model for the industry has moved from "one price, one service" games to one where games grow and evolve incrementally.

Financially, it is less expensive to create updates than a new game. The GaaS model not only improves the game by bringing in new ideas from the users, but it also lowers production costs, since it is concerned only with updates to an already functioning game. Additionally, the regular deployment of patches and updates can create a steady revenue stream for developers and publishers (NoBlue, 2018).

GaaS has also revolutionized the way gamers play. The survey data indicate that most respondents like the idea of GaaS (Exhibit 11). Players felt that they were listened to and were able to request new content, and they liked that GaaS enabled them to play their favorite game for as long as they wanted (Exhibit 7). The business model for the industry has moved from "one price, one service" games to one where games grow and evolve incrementally. GaaS means that some game franchises are kept alive for years, and players can build and develop their favorite characters over the long term. Even though gamers may buy fewer games, they will play them for longer periods of time and, in doing so, contribute to a steady revenue stream through microtransactions and DLCs. Additionally, gamers who become immersed in one game tend to build communities that grow and to share their common experiences, fostering a sense of belonging and loyalty to the product.

EXHIBIT 11 What do you think about GaaS video games?



Total number of respondents: 217.

Finally, microtransactions give players the option of customizing their gaming experience. Those who are willing to pay extra to instantly advance or to alter or adapt the appearance of their characters can do so. Microtransactions also give players the opportunity to compensate for their lack of experience or ability (Tomić, 2019). In short, microtransactions provide an avenue for games to be tailored to the evolving wishes of those who play them, whenever they want (Exhibit 6).

6.1 | Is the GaaS business model sustainable?

GaaS has been a game changer in the games industry, but at what cost? With the emergence of this new business model, players' expectations have risen: They now want more content and a better gaming experience for less money.

Microtransactions allow players to trade money for gaming time, and can destroy any sense of skills-based competition.

This has encouraged some developers and publishers to push boundaries and make what could be perceived as false claims for their games. For example, it was impossible to play EA's entire Battlefront II game for free; to enjoy all its features, it was necessary to pay at some point. Such requirements strengthen the view among gamers that certain publishers are only concerned with making money. Although most gamers understand that games developers and publishers are commercial enterprises, a game's perceived value is critical to its success. As illustrated in Exhibit 8, cost is the main reason why a gamer stops playing a game.

For some gamers, the revenue-generation aspects of the GaaS business model seem exploitative. For example, several games have slowly drifted into a "pay-to-win" mode (Moeller, 2018), which gives an advantage to players with greater purchasing power. Microtransactions allow players to trade money for gaming time, and can destroy any sense of skills-based competition. Players can skip some parts of the game and reach the top directly, but ultimately will know less about it than those who spend the time grinding to discover all its secrets (Makuch, 2016). In addition, such tactics may demotivate top players who face unfair competition from players who have paid to win (Klubnikin, 2018).

Related to these concerns is the debate concerning whether opting for loot boxes should be considered gambling. It has been shown that loot boxes have the capacity to manipulate players' behavior and can result in a significant outlay of time and money. In the long term, they can leave players with a sense of regret and frustration (Wiltshire, 2017).

7 | BEYOND GAMING

Looking at the experiences of the gaming industry, arguments can be made both for and against GaaS. Extrapolating beyond gaming, there are three key implications for other business models based on SaaS.

Offer minimum viable service to foster goodwill—It seems clear that a software service needs to provide satisfaction for its users if it is to be used. The key point here concerns building a minimum viable service (MVS)—that is, a service with enough features to satisfy enough customers to justify its future service development.

Although free-to-play games are readily available and allow gamers to play without having to buy anything, most GaaS games ultimately require some form of payment if they are to be enjoyed to the full, and this can leave some players feeling cheated. When a games company first launches an "unfinished" service, it needs to reach its MVS quickly to avoid losing users' goodwill. Similar considerations apply to SaaS, particularly when launching a new service. For example, when Dropbox, a cloud-based file storage and file-sharing app, was launched, it needed to offer enough functionality to meet the needs of users without driving them away *before* converting them into paying customers.

Ensure transparent cost structures to maintain existing customers—For SaaS to work, there needs to be a transparent cost structure that clearly states the level of investment needed to fully benefit from the software.

The results of this survey showed that some gamers did not appear to fully understand what they were getting for their money when they made microtransactions or bought DLCs. In the same way that failing to provide MVS in a new offering can turn away users before they become paying customers, failure to provide a transparent pricing structure can turn away existing customers. For example, Microsoft Office 365 has been criticized for having hidden costs for companies that migrate from standalone Office applications to the cloud-based Office 365.

Combine monetization practices for profitability—A final lesson for SaaS can be learned from the way in which game developers have found various ways to monetize the GaaS business model. Game companies extract revenue using relatively conventional methods, such as microtransactions, advergaming, subscription services, and season passes, as well as more unconventional approaches, such as loot boxes.

The challenge for SaaS business models is finding ways to balance "freemium" services with the need to generate a steady income stream so the organization can be profitable. Google Apps, such as Google Maps, Google

Play, and Hangouts, are well known for the way in which they are able to generate revenue by taking advantage of cross-linkages between different apps. This study of GaaS-based games, however, shows that multiple income streams can be created within a single application.

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AUTHOR BIOGRAPHIES

Fanny Vaudour is a gamer working in project management in the Paris studio of Ubisoft, a video games company. She graduated from KEDGE Business School's Grande École Program, where she researched the use of games-as-a-service business models. She can be reached at fmvaudour@gmail.com.

Aleksej Heinze, PhD, is an associate professor of digital marketing at KEDGE Business School, Marseille, France. He received his doctoral degree at the University of Salford, UK. His research interests center on digital marketing, with a particular focus on digital transformation and the use of search and social media. He can be reached at aleksej.heinze@kedgebs.com.

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